2025 Cryptopuzzle

Complete Four-Part Walkthrough

Compass substitution -> Atbash and Beaufort -> book cipher -> Braille and Atbash

Part 1 maritime map and compass puzzle

1. Puzzle Structure and General Method

The challenge is a chained cryptographic puzzle. Each protected PDF contains both a hidden message and the method needed to derive the password for the next stage. This guide is arranged as a teaching walkthrough: each section explains how to recognize the technique, provides a yellow Try it yourself pause, and then reveals the verified solution.

Stage	Primary technique	Goal
Part 1	Directional extraction + substitution	Derive the password for Part 2
Part 2	Atbash + Beaufort	Derive the password for Part 3
Part 3	Atbash + book cipher	Derive the password for Part 4
Part 4	Six-dot Braille + Atbash	Recover the scoreboard passphrase

Password formatting rule

Whenever a password is requested, use uppercase letters and remove spaces unless the puzzle explicitly says otherwise.

2. Part 1 - Compass-Derived Substitution Alphabet

The Latin instruction panel and scale beneath it

The Latin instruction is the operational clue. It tells the solver how to traverse the four arms of the compass and how to format the resulting password. A practical translation is:

Translated instruction

Construct the key in this order: North from top to bottom; South from bottom to top; East from right to left; West from left to right. Use the resulting key to unlock the next part, with all letters uppercase and no spaces.

Step 1: Read each compass arm in the specified direction

The letters are not meant to be read in ordinary clockwise order. Each cardinal direction has its own traversal direction. Use the table below so the reading direction and extracted letters remain visually aligned.

Compass arm	Required reading direction	Extracted letters
North	Top to bottom	Y U Q M I E A
South	Bottom to top	W S O K G C
East	Right to left	Z V R N J F B
West	Left to right	X T P L H D

Step 2: Concatenate the four sequences

Place the sequences together in the order named by the Latin text.

```
YUQMIEAWSOKGCZVRNJFBXTPLHD
```

Step 3: Use the result as a cipher alphabet

Write the ordinary alphabet above the extracted alphabet. To decrypt, find a ciphertext letter in the lower row and read the corresponding plaintext letter from the upper row.

```
Plain:  ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher: YUQMIEAWSOKGCZVRNJFBXTPLHD
```

Step 4: Decode the curved map text

The hidden ciphertext is:

```
HYPXZSKVVBIZYHS
```

TRY IT YOURSELF

Use the two-row alphabet above to decrypt the string. Record your proposed password before continuing.

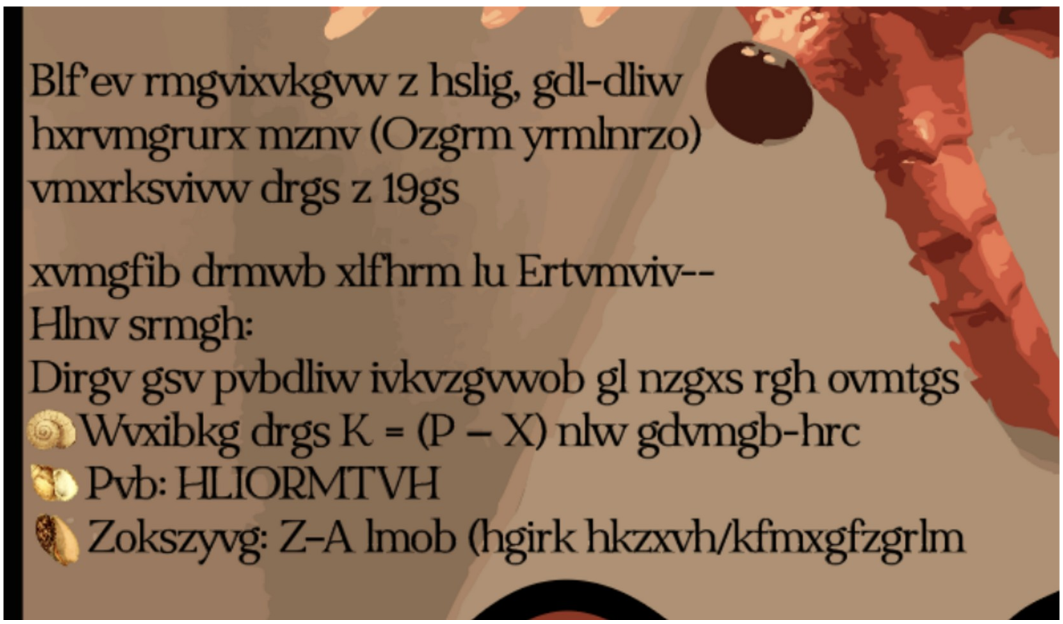
Write down your result before turning the page.

A few character mappings demonstrate the process: H maps to Y, Y maps to A, and P maps to W. Continuing across the entire string gives:

```
HYPXZSKVVBIZYHS  ->  YAWUNIKOOTENAYI
```

Part 2 password
YAWUNIKOOTENAYI

3. Part 2 - Atbash Instructions and Beaufort Decryption



Part 2 instruction text

Part 2 uses two layers. The first layer conceals the instructions; the second layer encrypts the actual password.

Step 1: Apply Atbash to the instruction text

Atbash reverses the alphabet. A becomes Z, B becomes Y, C becomes X, and so on. Because Atbash is reciprocal, the same operation both encrypts and decrypts.

```
ABCDEFGHIJKLMNOPQRSTUVWXYZ
ZYXWVUTSRQPONMLKJIHGFEDCBA
```

The decoded instruction reads, in substance:

Decoded Part 2 instruction
You have intercepted a short, two-word scientific name - a Latin binomial - enciphered with a 19th-century windy cousin of Vigenere. Decrypt using P = (K - C) mod 26. Key: SORLINGES.

```
Key: SORLINGES
Formula: P = (K - C) mod 26
```

Step 2: Recognize the Beaufort cipher

The formula P = (K - C) mod 26 is the defining arithmetic for Beaufort decryption when letters are numbered A=0 through Z=25.

Step 3: Repeat and align the key with the ciphertext

The key SORLINGES is shorter than the ciphertext. A polyalphabetic cipher needs one key letter for every ciphertext letter, so repeat SORLINGES from the beginning until both rows have the same length. The repetition does not create a new key; it simply cycles through the same nine key letters for each successive ciphertext position.

```
Ciphertext: ROYEKKCJKYWPLQKGCZUDXT
Key:        SORLINGESSORLINGESSORL
```

Step 4: Calculate each plaintext letter

Number the alphabet A=0 through Z=25 and evaluate P = (K - C) mod 26 at each position. Modulo 26 means that a negative result wraps forward into the 0-25 range.

```
Example 1 - first character:
Key S = 18, Cipher R = 17
P = (18 - 17) mod 26 = 1 = B

Example 2 - third character (wrap-around):
Key R = 17, Cipher Y = 24
P = (17 - 24) mod 26 = -7 mod 26 = 19 = T
```

TRY IT YOURSELF

Complete the Beaufort calculation for every position. Join the plaintext letters, then compare your result with the solution on the next page.

Write down your result before turning the page.

```
ROYEKKCJKYWPLQKGCZUDXT
BATHYDEVIUSCAUDACTYLUS
```

The result separates into the binomial scientific name Bathydevius caudactylus.

Part 3 password
BATHYDEVIUSCAUDACTYLUS

4. Part 3 - Atbash-Guided Book Cipher

Part 3 again uses Atbash to hide the instructions. Decoding the instruction block gives the following operational guidance:

Decoded Part 3 instruction
Use the hints on this page to deduce the cipher needed to decode the content. Go by the page number at the bottom of the reader, not the number written in the textbook. Ignore titles and headings when counting; focus on the main content on the page or paragraphs. Hint: Page / Word / Letter.

Coordinate format
Interpret every triple as Page / Word / Letter. For example, [165, 7, 3] directs you to reader page 165, the seventh body-text word, and the third letter of that word.

Step 1: Open the historical scan linked by the challenge

Use the reader page count shown by the online viewer, rather than a page number printed on the historical page itself.

Step 2: Apply consistent counting rules

Exclude headings such as PREFACE and EXTRACT. Begin with the first word of the body text. Count left to right and top to bottom. Punctuation does not create a new word. When a line-break hyphen divides one printed word, treat it as one word.

Step 3: Evaluate every coordinate in order

For each triple [page, word, letter], go to the specified page, find the specified word, and take the requested letter. Do not reorder the coordinates.

```
[2,6,1] [3,7,1] [4,1,4] [4,1,3]
[165,7,3] [168,2,5] [195,2,6] [7,5,1]
[162,5,3] [166,1,2] [175,3,2] [5,2,7]
[207,10,4] [13,1,5] [195,3,1] [6,2,10]
```

```
[7,3,2] [175,8,2] [9,1,2] [167,1,1]
[178,4,3] [176,2,3]
```

TRY IT YOURSELF

Work through the coordinates in order using the historical scan linked by the challenge. Record one letter per coordinate and join the letters before continuing.

Write down your result before turning the page.

Step 4: Concatenate the extracted letters

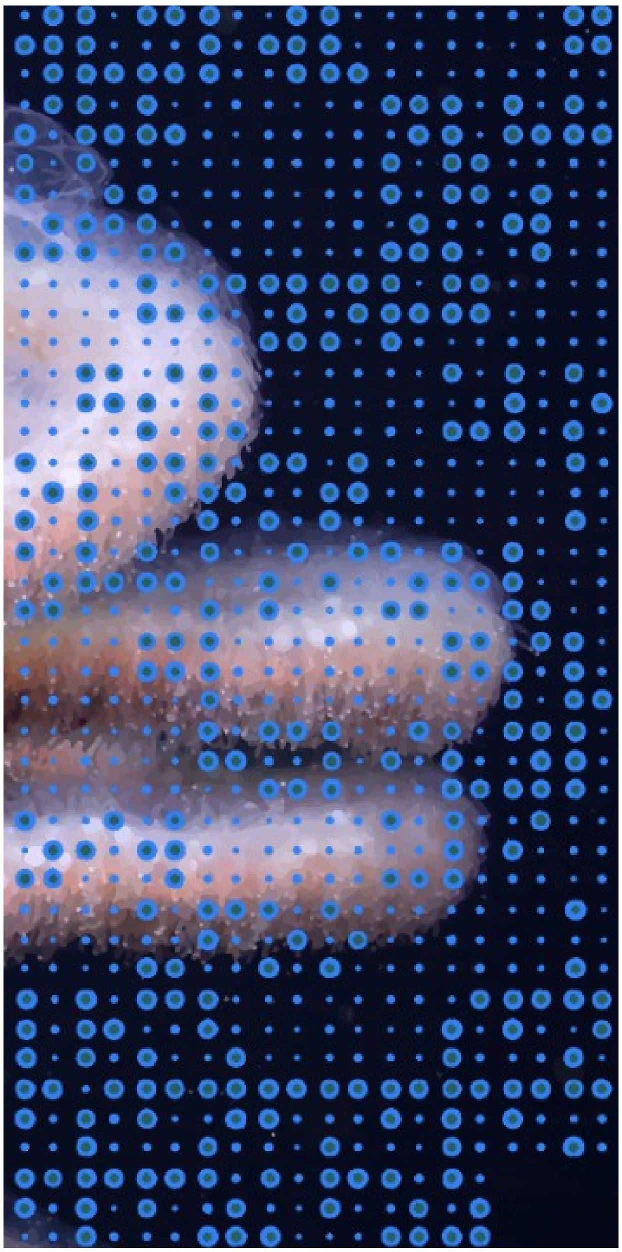
The 22 letters combine to form:

```
PRAYADUBIASIPHONOPHORE
```

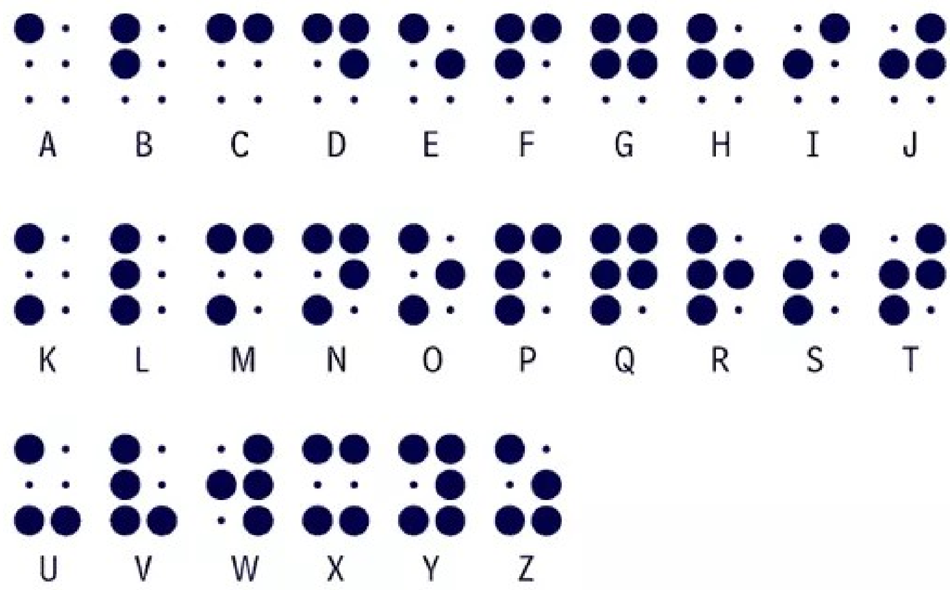
This can be read as PRAYA DUBIA SIPHONOPHORE, which fits the marine-organism theme and provides a strong semantic verification that the extraction is correct.

Part 4 password
PRAYADUBIASIPHONOPHORE

5. Part 4 - Braille Grid and Final Atbash Layer



Part 4 image with a regular grid of large rings and small dots



Six-dot Braille alphabet key used to decode the Part 4 overlay

The blue overlay is highly regular: every position contains either a large hollow ring or a small dot. This binary distinction is the clue that the overlay encodes Braille.

Step 1: Assign the two visual states

Treat a large hollow ring as a raised Braille dot. Treat a small dot as an unraised or blank Braille position.

Step 2: Divide the grid into Braille cells

Standard literary Braille uses six positions arranged in two columns by three rows:

```
1 4
2 5
3 6
```

Starting at the upper-left of the overlay, divide the pattern into adjacent 2-column by 3-row cells. Read cells from left to right, then continue on the next row of cells.

Step 3: Translate each six-position pattern

For each cell, note which numbered positions contain large rings. Match that set of positions to the Braille alphabet reference. Spaces or punctuation appear as blank or special cells depending on the pattern.

Step 4: Apply Atbash to the Braille-derived letters

The first Braille transcription remains alphabetically reversed. Apply Atbash one final time to obtain readable English.

TRY IT YOURSELF

Decode each 2-column by 3-row Braille cell using the supplied key. Then apply Atbash to your transcription before continuing.

Write down your result before turning the page.

Decoded final instruction

DECODE THIS MESSAGE AND SUBMIT THE FOLLOWING PASSPHRASE TO THE SCOREBOARD FOR FULL POINT: HUMUHUMUNUKUNUKUAPUAA

Final scoreboard passphrase

HUMUHUMUNUKUNUKUAPUAA

End of walkthrough